

Response to the Department of Health and Human Services Request for Information: Accelerating the Adoption and Use of Artificial Intelligence as Part of Clinical Care

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Helios Intelligence

I. Introduction

Helios respectfully submits these comments in the attached response to the Department of Health and Human Services (HHS) Request for Information (RFI) on accelerating the adoption and use of Artificial Intelligence (AI) as part of clinical care, published in the Federal Register on December 23, 2025 (90 FR 60108).

Helios is a clinical AI company building advanced models and specialized agents to automate and enhance clinical care and biomedical research. Our platform unifies fragmented, multimodal patient data — from EHRs, labs, genomics, imaging, and wearables — into a continuously updated, longitudinal patient model that powers clinical reasoning, decision support, and care optimization. We work across the healthcare ecosystem, from health systems and clinical laboratories to payers and direct-to-consumer health platforms.

We applaud HHS for soliciting industry input on this critical topic. Our comments attached are grounded in our direct experience building, deploying, and validating clinical AI solutions across the healthcare ecosystem.

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II. Responses to Specific Questions

Question 1: What are the biggest barriers to private sector innovation in AI for healthcare and its adoption and use in clinical care?

The single greatest barrier to AI innovation in healthcare is the **fragmentation of patient data across incompatible systems and modalities**. Patient information today is scattered across EHRs, laboratory information systems, genomic databases, imaging archives, pharmacy records, and wearable devices — each with its own format, access model, and update cycle. No common data representation exists across these modalities, meaning each requires specialized parsing and interpretation.

This fragmentation has a cascading effect on both clinical care and AI development. Clinicians are forced to serve as the manual "integration layer," synthesizing numerous data sources per patient encounter — an inefficient process that leads to missed connections and diagnostic delays. For AI developers, the lack of standardized, interoperable data makes it extraordinarily difficult and expensive to build models that can reason holistically about a patient's health over time. Even the most advanced large language models are not natively built to route each dataset to the right analytical tool or to

maintain a longitudinal patient state, and as a result, their clinical reasoning degrades in multimodal contexts.

Beyond data fragmentation, additional barriers include **regulatory uncertainty** around the classification and oversight of non-medical-device AI tools, the **absence of reimbursement pathways** for AI-enabled clinical services, and the **difficulty of integrating** new AI tools into existing clinical workflows without disrupting care delivery. The reimbursement landscape is particularly challenging: as of January 2026, only three AI solutions have received Category I CPT codes, and most AI tools are billed under temporary Category III codes with no guaranteed payment. This uncertainty limits the financial viability of clinical AI solutions and incentivizes developers to prioritize administrative AI—where revenue models are more predictable—over clinical AI with greater patient impact.

Question 2: What regulatory, payment policy, or programmatic design changes should HHS prioritize to incentivize the effective use of AI in clinical care and why?

HHS should prioritize a multi-pronged approach that combines regulatory clarity, payment innovation, and programmatic support. We propose several frameworks that we believe could be transformative.

Regulatory and Programmatic Priorities:

Government-Sponsored Clinical AI Agent Marketplace. HHS should establish a curated, government-endorsed platform where validated clinical AI agents can be discovered, evaluated, and deployed by health systems. Health systems today face an overwhelming and opaque landscape of AI tools with no standardized way to compare them. A government-endorsed marketplace, with clear validation requirements and transparent performance data, would dramatically lower the barrier to adoption. Agents could be organized by clinical specialty, condition (e.g., cardiology agents, chronic care agents, oncology agents), and risk level, allowing health systems to find the right tool for their needs. The ARPA-H ADVOCATE program, which is funding the development of an FDA-authorized AI agent for cardiovascular disease, demonstrates that the government is already investing in condition-specific AI agents; the marketplace concept would scale this approach across the full spectrum of clinical care.

NPI-Equivalent Identifiers for AI Agents. Just as every clinician in the United States has a unique National Provider Identifier, validated AI agents should be assigned unique identifiers that allow them to be tracked, credentialed, audited, and billed against within the healthcare system. This would enable healthcare organizations to establish **AI Agent Credentialing Committees** — formal governance bodies, similar to existing medical staff committees, that review, approve, and continuously monitor the AI agents deployed in their clinical environments. This framework would bring the same rigor and accountability to AI deployment that already exists for human clinicians, building trust among providers and patients alike.

Expanded ARPA-H Funding for Disease-Specific Agents. The ADVOCATE model — which funds the development of autonomous AI agents with a clear FDA authorization pathway and a defined timeline — should be replicated for other high-burden conditions such as diabetes, COPD, and cancer. This would accelerate the creation of a library of validated, condition-specific AI agents that could populate the proposed marketplace.

Payment Policy Priorities:

Expand the CMS ACCESS Model. The CMS ACCESS (Advancing Chronic Care with Effective, Scalable Solutions) model is the right foundation — it already tests outcome-aligned payment in Original Medicare and explicitly promotes technology-enabled care. HHS should make ACCESS the rule, not the exception, by rapidly expanding the model beyond its current scope and using it as the template for how AI-enabled care is reimbursed across Medicare. Specifically, ACCESS should be extended to cover a broader range of AI-driven interventions — not just chronic care management, but also AI-powered diagnostics, care gap detection, and preventive screening — so that any AI tool demonstrating measurable improvements in patient outcomes (reduced readmissions, earlier detection, lower total cost of care) has a clear path to reimbursement.

Accelerate Development of CMAA Codes. Beyond expanding ACCESS, HHS should work with the AMA to accelerate the development of the proposed "Clinically Meaningful Algorithmic Analyses" (CMAA) codes — a new coding classification within the CPT code set specifically designed for algorithm-based services. Today, most clinical AI tools are limited to temporary Category III codes with no guaranteed payment and wide geographic variation in pricing set by regional MACs. CMAA codes would create a dedicated, nationally consistent billing pathway for clinical AI, giving developers the reimbursement certainty they need to invest in clinical — not just administrative — AI.

AI-Specific Value-Based Purchasing Incentives. Finally, HHS should explore AI-specific value-based purchasing incentives that go beyond ACCESS. For example, Medicare Advantage Star Ratings could be updated to reward plans that deploy validated AI agents for proactive care management, and shared savings programs could include explicit bonuses for AI-driven cost reductions that maintain or improve quality.

Question 3: For non-medical devices, what novel legal and implementation issues exist... and what role, if any, should HHS play to help address them?

The deployment of non-medical-device AI in clinical care raises several novel legal and implementation challenges that existing governance structures are not fully equipped to address.

Liability and accountability are the most pressing concerns. When an AI agent contributes to a clinical decision — for example, by flagging a care gap, interpreting a lab result, or synthesizing a patient's medical history — the question of who bears responsibility for errors is often unclear. Is it the AI developer, the healthcare provider who acted on the recommendation, or the institution that deployed the tool? HHS can play a constructive role by developing **model frameworks for shared responsibility** that encourage transparency, accountability, and the use of validated AI tools, without imposing liability structures so burdensome that they stifle innovation. The AI Agent Credentialing Committees proposed in our response to Question 2 would further clarify lines of accountability by creating a formal institutional process for evaluating and approving AI agents before deployment.

Privacy and security concerns are amplified by the volume and sensitivity of data that clinical AI systems process. While HIPAA provides a critical baseline, the emergence of AI tools that synthesize data across multiple modalities and maintain longitudinal patient states requires a renewed focus on data governance. HHS should promote the adoption of **privacy-enhancing technologies** such as federated learning and differential privacy, and provide explicit guidance on how existing HIPAA provisions apply to AI-specific data processing activities, including the generation and storage of AI-derived patient representations.

Implementation challenges include the need for robust change management within healthcare organizations, the difficulty of integrating AI tools with legacy IT systems, and the absence of standardized credentialing or certification processes for clinical AI. The NPI-equivalent identifier and

credentialing committee frameworks proposed in Question 2 would directly address the credentialing gap and give health systems a structured pathway for governing AI deployment.

Question 4: For non-medical devices, what are the most promising AI evaluation methods... and should HHS further support these processes?

The most promising evaluation methods are those that assess AI performance in realistic clinical contexts using clinician-led, blinded, randomized protocols. Standard NLP benchmarks often fail to capture the nuances of clinical decision-making. Evaluations should test models on their ability to perform specific, clinically relevant tasks — such as synthesizing multimodal patient data, identifying care gaps, or generating accurate differential diagnoses — as judged by practicing physicians.

However, expert physician evaluation does not scale. To solve this, HHS should fund the creation of a National Repository of Clinician-Backed Grading Rubrics — a structured, open-access library of evaluation criteria authored and validated by hundreds or thousands of expert physicians across every major specialty. These rubrics would codify what "good" clinical reasoning looks like for specific tasks (e.g., interpreting a complex lab panel, triaging disease presentation, recommending a treatment pathway for a comorbid patient). Once established, these rubrics could be used to power LLM-as-a-Judge evaluation systems — where a secondary AI model, guided by the physician-authored rubrics, can grade the outputs of clinical AI tools at scale with a level of rigor that adheres to strict guidelines set by the expert community. This approach would make high-quality, clinician-aligned evaluation continuous, scalable, and affordable, rather than limited to expensive, one-off human review studies.

For post-deployment evaluation, continuous monitoring frameworks that track model performance, drift, and safety in real-world clinical settings are essential. The ARPA-H ADVOCATE program's plan to develop a disease-agnostic supervisory agent for monitoring AI systems is a promising approach that could be generalized across the industry.

HHS should further support these processes through several mechanisms:

Federated Evaluation Networks. HHS should fund the creation of government-supported infrastructure that allows AI models to be tested across multiple institutions on diverse patient populations without sharing raw patient data. This infrastructure would preserve privacy while enabling robust, multi-site validation that produces generalizable results.

Open-Source Clinical AI Benchmarks. HHS should fund the creation of publicly available, high-fidelity synthetic patient datasets and standardized evaluation toolkits for a range of clinical specialties. This would level the playing field for all developers — including smaller, innovative companies — and ensure that models are benchmarked against common, clinically meaningful standards rather than proprietary datasets.

Prize Competitions. HHS should sponsor prize competitions that benchmark AI models on specific, high-impact clinical challenges, such as care gap detection and treatment pathway optimization. These would stimulate innovation, establish public performance baselines, and generate valuable data on the state of the art.

Question 5: How can HHS best support private sector activities... to promote innovative and effective AI use in clinical care?

HHS can most effectively support the private sector by acting as a **convener, endorser, and data provider**.

As a convener, HHS should bring together multi-stakeholder working groups — including AI developers, healthcare providers, payers, patient advocates, and academic researchers — to develop consensus-based standards for clinical AI evaluation, deployment, and monitoring. These standards should be flexible enough to accommodate the rapid pace of AI innovation while rigorous enough to ensure patient safety.

As a data provider, HHS should make more **de-identified government datasets available** for AI model training and evaluation. The NIH, CMS, and VA hold vast repositories of clinical data that, if made accessible through secure, privacy-preserving mechanisms such as federated learning infrastructure, could dramatically accelerate the development of high-quality clinical AI models. The transformative potential of accessible research data for training multimodal clinical models cannot be overstated.

As an endorser, HHS should lend its credibility to industry-led certification and testing initiatives that meet rigorous standards. The **CMS Health Tech Ecosystem pledge program** for conversational AI assistants, which already includes early adopters like major technology companies and health systems, is an excellent model for how government can catalyze AI adoption through voluntary industry commitments without heavy-handed regulation.

Question 6: Where have AI tools deployed in clinical care met or exceeded performance and cost expectations and where have they fallen short? What kinds of novel AI tools would have the greatest potential to improve health care outcomes, give new insights on quality, and help reduce costs?

Where AI has met or exceeded expectations:

AI tools have demonstrated exceptional performance in **clinical documentation and EHR analysis**, significantly reducing the time clinicians spend on administrative tasks and directly addressing provider burnout. AI-driven **care gap detection** has also shown promising results, with intelligent agents scanning patient records against clinical guidelines to flag missed screenings, lapsed medications, and overdue follow-ups — enabling proactive intervention at scale. In the payer space, AI is increasingly being used for **prior authorization automation**, with some organizations targeting near-real-time decision-making.

Where AI has fallen short:

AI tools have underperformed when deployed as generic, one-size-fits-all solutions without domain-specific training or clinical validation. General-purpose LLMs, while impressive in breadth, often lack the depth of clinical reasoning required for safe and effective use in healthcare. They can hallucinate, miss critical context, and fail to account for the multimodal, longitudinal nature of patient data. The industry has also seen troubling examples of AI being used to automate insurance claim denials without adequate clinical oversight, eroding public trust and underscoring the need for rigorous governance.

Novel AI tools with the greatest potential:

The tools with the greatest transformative potential are those that enable dynamic, computable models of individual human physiology — what we term "**World Simulations of Humans.**" This approach represents a paradigm shift beyond static data aggregation towards creating a dynamic, computable model of an individual's entire physiology. Such a simulation would serve as the **standard platform for**

integrating all multimodal data — from genomics and proteomics to imaging, EHR data, and real-time wearable streams. By fusing these sources into a unified, dynamic model, these simulations would enable predictive modeling of disease progression, *in silico* simulation of individual treatment responses, and truly personalized care at a biological level. HHS should consider direct investment in the foundational research and infrastructure for these models, as they represent the ultimate platform for preventive medicine and long-term cost reduction.

Additionally, **condition-specific autonomous AI agents** — like the cardiovascular agent being developed through ARPA-H's ADVOCATE program — represent a paradigm shift from decision-support tools to active participants in care delivery. Expanding this model to other disease areas would be particularly impactful.

Question 7: Which role(s), decision maker(s), or governing bodies within health care organizations have the most influence on the adoption of AI for clinical care? What are the primary administrative hurdles?

The most influential decision-makers for AI adoption are Chief Medical Officers (CMOs), Heads of Innovation, and increasingly, dedicated AI leads within healthcare organizations. These leaders are closest to the clinical pain points and are often the ones pushing to accelerate beyond what their organizations can build in-house. They recognize that the pace of AI development in the broader market far exceeds what any single health system's internal team can match, and they are actively seeking external partners and tools. C-suite executives (CEO, CFO, CIO) make the final investment decisions, primarily motivated by ROI, regulatory compliance, and competitive positioning, while clinical department chairs and physician leaders have outsized influence on whether frontline clinicians will actually adopt and trust a new tool in practice.

The primary administrative hurdles are evolving. Most healthcare organizations have focused initial AI efforts on automating the operational basics: scheduling, billing, claims processing, and documentation. These segments are important, but they represent the floor, not the ceiling. The larger challenge — and the larger opportunity — is shifting AI adoption toward higher-value clinical applications: identifying incorrect billing codes before they cause downstream problems, detecting care gaps where patients should be receiving screenings or treatments they are not, reducing misdiagnosis through AI-assisted clinical reasoning, moving toward preventive health by identifying at-risk patients before acute events occur, improving patient education and engagement through personalized AI-driven outreach, enabling payers to perform real-time risk tracking and simulation across their populations, and dramatically compressing the time it takes to assemble care plans for complex disease patients — including finding the right specialists, trials, and resources nationwide.

The AI Agent Credentialing Committees proposed elsewhere in this response would help organizations navigate this transition by providing a formal, standardized governance framework for evaluating, approving, and monitoring AI agents across both operational and clinical use cases — giving innovation leaders and CMOs a clear process to move faster with confidence.

Question 8: Where would enhanced interoperability widen market opportunities, fuel research, and accelerate the development of AI for clinical care?

Enhanced interoperability is the single most important technical enabler of clinical AI at scale. The universal adoption and enforcement of **FHIR-based APIs** would transform the healthcare AI landscape.

Market Opportunities

Universal adoption and enforcement of FHIR-based APIs would create a true plug-and-play ecosystem for clinical AI tools. Today, a significant portion of development time and cost is consumed by custom integrations with each client's unique data infrastructure. If all EHRs, lab systems, and payer platforms adhered to common data standards, AI developers could build once and deploy everywhere — dramatically lowering costs, accelerating time-to-market, and enabling smaller, innovative companies to compete with entrenched incumbents.

This would be particularly impactful for **AI-driven prior authorization automation**. With CMS rules already mandating electronic prior authorization and faster decision timelines between 2026 and 2027, a fully interoperable system would allow AI agents to securely compile necessary clinical documentation and submit it to payers in real time, drastically reducing administrative burden and care delays. The same interoperability would enable AI agents to seamlessly access and synthesize data across providers, creating the unified patient view that is essential for high-quality clinical reasoning.

Research and AI Development

Interoperability alone is necessary but not sufficient. Even with standardized data formats, it remains extraordinarily difficult for AI developers and researchers to access large-scale, longitudinal datasets — spanning EHRs, lab results, imaging, and genomics — and train models on them at scale. Privacy concerns, institutional data-sharing restrictions, and the sheer cost of assembling and de-identifying these datasets create a bottleneck that interoperability standards alone cannot solve.

HHS should address this directly through two complementary approaches. First, HHS should establish a National Clinical AI Training Repository — a secure, centralized (or federated) data environment with the necessary compute resources where vetted researchers and developers can train models on large-scale, longitudinal, multimodal clinical data under strict privacy and governance controls. Second, HHS should invest in the generation and public release of high-fidelity synthetic clinical datasets built on top of real-world data. Synthetic data that faithfully preserves the statistical properties, clinical patterns, and longitudinal complexity of real patient records — without exposing any individual's information — would democratize access to training data and dramatically accelerate the development of clinical AI across the entire ecosystem.

Without one or both of these interventions, only a handful of organizations with privileged data access will be able to build frontier clinical AI models — limiting competition, innovation, and ultimately patient benefit.

The areas where enhanced interoperability would have the greatest impact include genomic data, wearable and remote monitoring data (which currently lacks any standardized format), lab results, and imaging — all linked to longitudinal EHR records.

Question 9: What challenges within health care do patients and caregivers wish to see addressed by the adoption and use of AI in clinical care? Equally, what concerns do patients and caregivers have?

Challenges patients and caregivers want AI to address:

The most fundamental challenge is the gap between the pace of scientific discovery and the personalization of care. Millions of peer-reviewed papers are published each year, but the vast majority of this knowledge never reaches the individual patient in a meaningful way. Patients want **personalized, science-grounded health guidance** — not generic recommendations, but insights tailored to their unique biology, medical history, and circumstances.

Patients and caregivers also want help **navigating complex medical journeys**, particularly for cancer and chronic conditions. AI agents that can synthesize multi-specialist findings, connect patients to relevant clinical trials, and provide a unified view of a complex medical history would be transformative.

Finally, patients want **proactive rather than reactive care**. They want to understand health risks before they become emergencies, and they want their healthcare providers equipped with tools that enable early identification and intervention.

Concerns patients and caregivers have:

The primary concerns are around **data privacy and security** — patients want to know that their most sensitive health information is protected and used only with their consent. There are also concerns about **algorithmic bias** and the risk that AI models trained on non-representative datasets could perpetuate or amplify existing health disparities. Finally, patients value the **human relationship** in healthcare and want assurance that AI will augment, not replace, the clinician-patient bond.

The CMS Health Tech Ecosystem pledge program for conversational AI assistants provides a useful model for addressing these concerns. By having companies publicly commit to building tools that are secure, transparent, and patient-centered, the program creates accountability and begins to build the public trust that widespread adoption will require.

Question 10: Are there specific areas of AI research that HHS should prioritize to accelerate the adoption of AI as part of clinical care?

HHS should prioritize research in the following areas:

Multimodal AI for Patient Representation and Digital Twins. The ability to create a unified, continuously updated model of a patient's health — integrating data from EHRs, genomics, proteomics, metabolomics, imaging, and wearables — is the foundation of next-generation clinical AI. Research that advances the science of Patient Digital Twins, including methods for fusing heterogeneous data types and generating rich molecular profiles from routine clinical data, should be a top priority.

Autonomous and Semi-Autonomous Clinical AI Agents. Building on the ARPA-H ADVOCATE model, HHS should fund research into the design, safety, and governance of AI agents that can autonomously perform clinical tasks — from care coordination and medication management to patient triage and monitoring. This funding would encompass research on supervisory AI systems that can monitor and ensure the safety of other AI agents in real time.

Implementation Science for Clinical AI. Understanding how to effectively integrate AI into clinical workflows — including the organizational, behavioral, and technical factors that determine success or failure — is as important as the AI technology itself. HHS should fund research on best practices for AI deployment, clinician training, change management, and the design of AI Agent Credentialing Committees.

AI Safety, Fairness, and Explainability. Research on methods to detect and mitigate bias in clinical AI models, to ensure equitable performance across diverse patient populations, and to make AI reasoning transparent and interpretable to clinicians and patients is essential for building and maintaining public trust.